St. Anne's Arts and Science College

Department of Computer Science

Team Members	Register No	Project Name	Phase Topic
JAISREE.R	222409465	FitTrack AI - Personalized Fitness Recommendations Powered by AI	Phase 2 REQUIREMENT ANALYSIS
JANU SRI.D	222409466		
JECHINTHA JOYCE.S	222409467		
JENITA.G	222409468		

PROJECT PHASE DOCUMENT

FITTRACK AI

Personalized Fitness Recommendations Powered by AI

PHASE 2

REQUIREMENT ANALYSIS

Project Name: AI FitTrack

CONTENTS

2.1	Purpose of Requirement Analysis
2.2	Software Requirements
2.3	Hardware Requirements
2.4	User Requirements
2.5	Functional Requirements
2.5.1	User Authentication
2.5.2	Workout Management
2.5.3	AI Recommendation
2.5.4	Fitness Insights
2.5.5	API and Error Handling
2.6	Data Requirements
2.7	External Service Requirement
2.8	Phase Summary

2.1 Purpose of Requirement Analysis

The requirement analysis phase identifies and organizes the technical and functional requirements needed to develop AI FitTrack. These requirements are derived from the proposed fitness-tracking services, user authentication mechanism, workout-management features, data-storage needs, and AI-powered services.

This phase considers the software environment, hardware resources, user needs, functional behaviour, data requirements, and external services necessary to implement the backend. Establishing these requirements provides a clear basis for the subsequent design and development activities.

2.2 Software Requirements

The following software platforms and development tools are identified for building and testing AI FitTrack:

Software / Tool	Purpose / Requirement
Operating System	Windows 10/11, macOS, or Linux
Node.js	Version 16 or above; JavaScript runtime for the backend
npm	Version 8 or above; package and dependency management
Express.js	Development of RESTful backend routes and services
MongoDB	Storage of user and workout information
Mongoose	MongoDB interaction, schemas, and data modelling

Visual Studio Code	Source-code editing and development environment
Postman	Testing REST API requests and responses

The project also uses supporting packages to provide configuration, security, cross-origin support, and development convenience:

- dotenv – loads application configuration from environment variables.
- bcryptjs – hashes passwords before they are stored.
- jsonwebtoken – creates and verifies JSON Web Tokens for authentication.
- cors – enables configured cross-origin requests.
- Nodemon – restarts the development server when source files change.

2.3 Hardware Requirements

A standard modern computer is sufficient for developing and testing the backend. The identified hardware requirements are summarized below:

Component	Requirement
Processor	Intel Core i5, 8th Generation or above, or AMD Ryzen 5
RAM	Minimum 8 GB; 16 GB recommended
Storage	At least 1 GB of available space

These specifications provide a suitable working environment for the development tools, backend server, MongoDB database, and API testing application. Actual resource usage may vary depending on the local setup and other applications running.

2.4 User Requirements

The primary user is an individual who has registered with AI FitTrack and authenticated successfully. The system is expected to allow the user to:

- Create an account through the registration facility.
- Log in using registered credentials.
- Access protected resources using authentication.
- Create and manage personal workout records.
- View previously recorded workouts and retrieve an individual record.
- Search workout records using available criteria.
- Update or delete existing workout records.
- Request personalized AI-based workout recommendations.
- Obtain fitness insights derived from workout statistics.

2.5 Functional Requirements

The functional requirements describe the principal services that the backend must provide. They are grouped below according to the system's major features.

2.5.1 User Authentication

The system must provide user registration and login facilities. Passwords must be protected using bcryptjs, and JSON Web Tokens (JWTs) must be used to authenticate requests to protected APIs. Requests without valid authorization must not be allowed to access protected resources.

2.5.2 Workout Management

Authenticated users must be able to manage workout records through RESTful API endpoints. The required operations are:

- Add a workout record.
- View workout records.
- Retrieve a specific workout.
- Update existing workout information.
- Delete a workout record.
- Search workouts by name, category, or date.

2.5.3 AI Recommendation

The backend must provide an AI-based recommendation service through Google Gemini. The recommendation process uses relevant user information, including age, fitness goal, and experience level, to generate personalized fitness guidance.

2.5.4 Fitness Insights

The backend must generate fitness insights from recorded workout information. The identified statistics include:

- Total number of workouts.
- Average workout duration.
- Total calories burned.

2.5.5 API and Error Handling

The application must follow a RESTful API structure and return responses in JSON format. Input validation, route protection, and centralized error handling must be incorporated to support consistent API behaviour and handle invalid, unauthorized, or otherwise unsuccessful requests appropriately.

2.6 Data Requirements

AI FitTrack requires two primary collections in MongoDB. Their intended contents are summarized in the following table:

Collection	Data Stored	Purpose
Users	Name, email, password hash, and timestamps	Stores account and authentication-related information.
Workouts	Workout name, category, duration, calories burned, workout date, and	Stores workout records linked to their owner.

	associated user	

Each workout is associated with a user through a user reference. This represents a one-to-many relationship: one user may have multiple workout records, while each workout record is linked to its associated user.

[INSERT DATA RELATIONSHIP / ER DIAGRAM HERE]

Figure 2.1: Proposed relationship between the Users and Workouts collections.

2.7 External Service Requirement

Google Gemini is required to support the application's AI-based features. The backend sends relevant fitness information to the Gemini API to obtain personalized recommendations and fitness insights. The Gemini API key must be stored in the application's environment configuration and must not be hard-coded directly into the source code.

2.8 Phase Summary

The requirement analysis phase identifies the software and hardware environment, user needs, functional services, data structures, and external AI dependency for AI FitTrack. The defined requirements establish the basis for designing a secure RESTful backend with user authentication, workout management, search, fitness insights, and Google Gemini-powered recommendations.